

PASTA worksheet

Stages	Sneaker company
I. Define business and security objectives	Make 2-3 notes of specific business requirements that will be analyzed. <i>Users should be able to create member profiles either directly within the application or by linking third-party accounts. Since the application handles financial transactions, it must follow PCI-DSS compliance requirements to ensure secure processing and protection of payment information.</i>
II. Define the technical scope	List of technologies used by the application: • <i>Application programming interface (API)</i> • <i>Public key infrastructure (PKI)</i> • <i>SHA-256</i> • <i>SQL</i> APIs play an important role in enabling communication and data sharing between customers, employees, and business partners, making them a high-priority area for security. Because APIs frequently process sensitive information and connect multiple systems, they increase the overall attack surface of an application. As a result, the specific APIs being used and their security configurations should be carefully evaluated to reduce potential vulnerabilities.
III. Decompose application	<u>Sample data flow diagram</u>
IV. Threat analysis	List 2 types of threats in the PASTA worksheet that are risks to the information being handled by the application. • <i>Injection</i> • <i>Session hijacking</i>
V. Vulnerability analysis	List 2 vulnerabilities in the PASTA worksheet that could be exploited. • <i>Lack of prepared statement</i> • <i>Broken API token</i>

VI. Attack modeling	Sample attack tree diagram
VII. Risk analysis and impact	List 4 security controls that you've learned about that can reduce risk. SHA-256, incident response procedures, password policy, principle of least privilege
